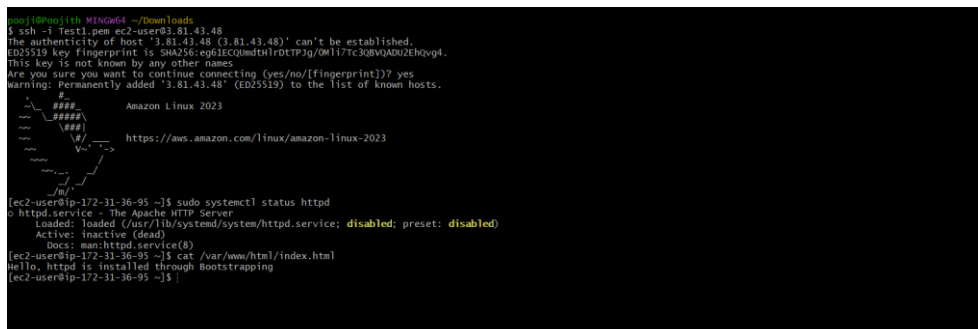
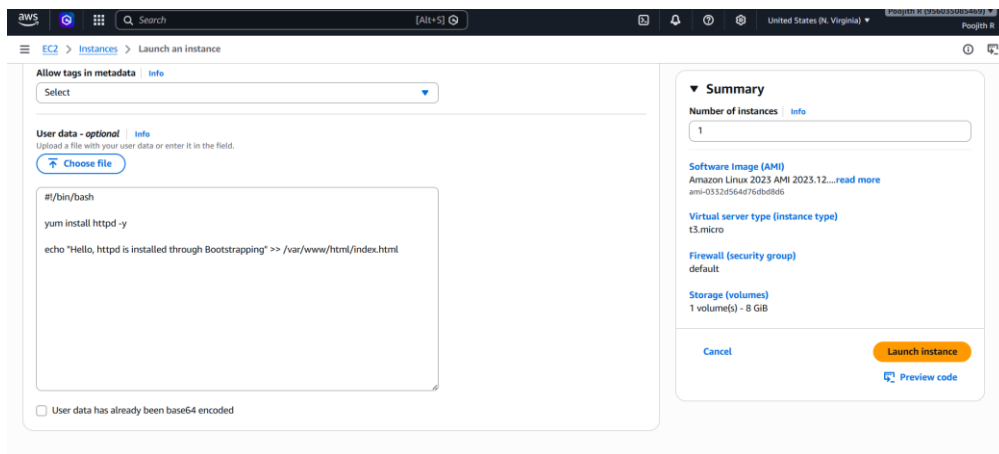
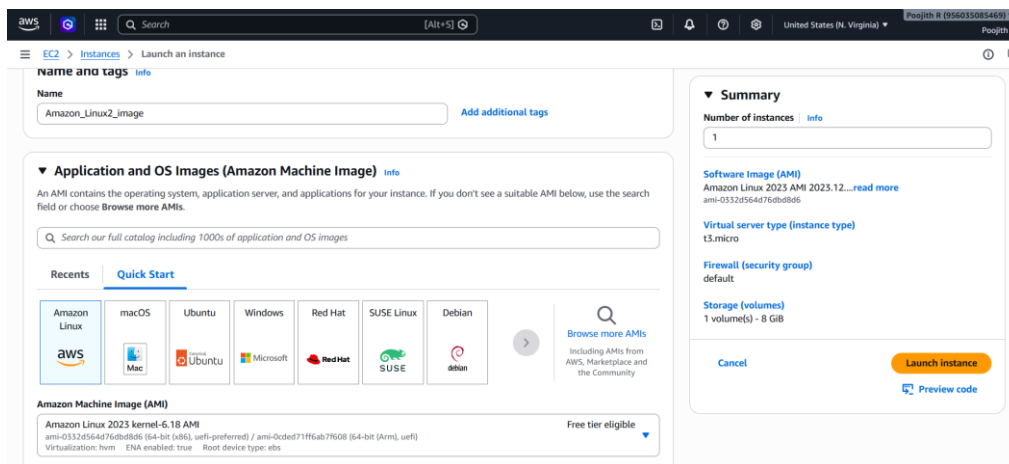


# Task on EC2

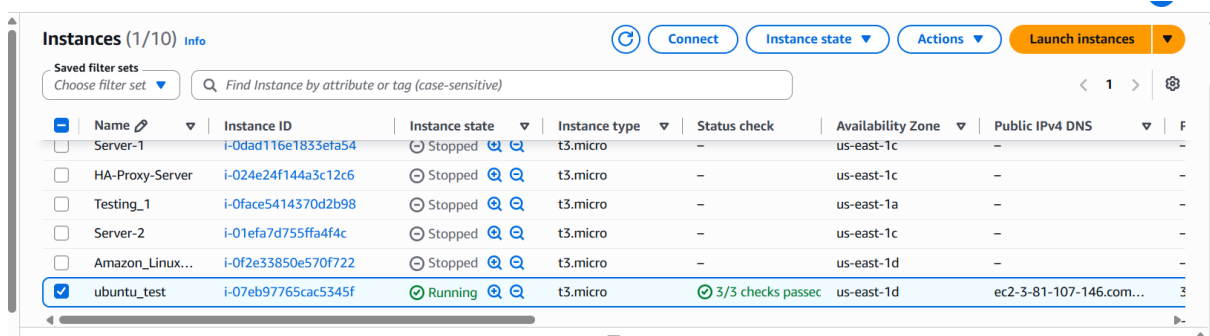
## 1. Launch one EC2 using Amazon Linux 2 image and add a script in user data to install Apache.

- ➔ Launched an EC2 instance using the Amazon Linux 2 AMI.
- ➔ Selected the required instance type, key pair, subnet, and security group.
- ➔ Added a user data script to install and start the Apache web server during instance launch.
- ➔ Connected to the EC2 instance using SSH.
- ➔ Verified that Apache was installed and running.
- ➔ Tested the Apache default page using the EC2 public IP.

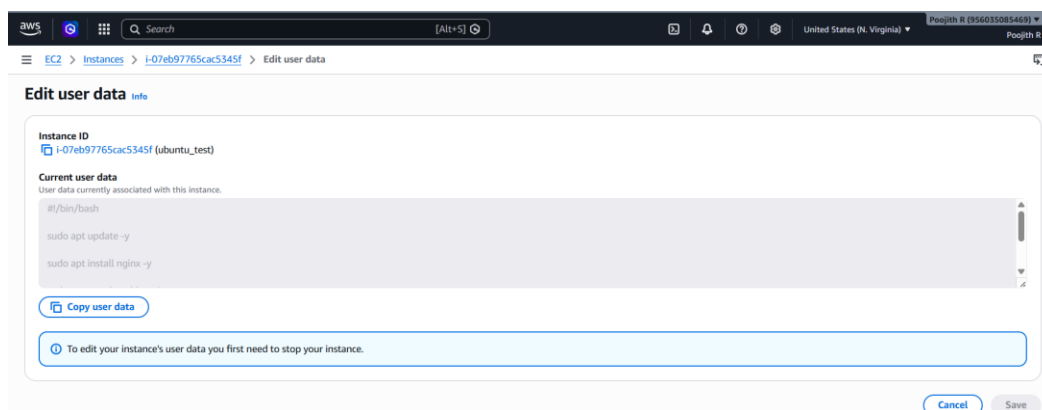


## 2. Launch one EC2 using Ubuntu image and add a script in user data to install Nginx.

- ➔ Launched an EC2 instance using the Ubuntu AMI.
- ➔ Configured the instance type, key pair, subnet, and security group.
- ➔ Added a user data script to install and start the Nginx web server automatically.
- ➔ Connected to the Ubuntu EC2 instance using SSH.
- ➔ Verified that Nginx was installed and running.
- ➔ Accessed the Nginx default page using the EC2 public IP.



| Name            | Instance ID         | Instance state | Instance type | Status check      | Availability Zone | Public IPv4 DNS         |
|-----------------|---------------------|----------------|---------------|-------------------|-------------------|-------------------------|
| Server-1        | i-0dad116e1833efa54 | Stopped        | t3.micro      | -                 | us-east-1c        | -                       |
| HA-Proxy-Server | i-024e24f144a3c12c6 | Stopped        | t3.micro      | -                 | us-east-1c        | -                       |
| Testing_1       | i-0face5414370d2b98 | Stopped        | t3.micro      | -                 | us-east-1a        | -                       |
| Server-2        | i-01efa7d755ffa4f4c | Stopped        | t3.micro      | -                 | us-east-1c        | -                       |
| Amazon Linux... | i-0f2e33850e570f722 | Stopped        | t3.micro      | -                 | us-east-1d        | -                       |
| ubuntu_test     | i-07eb97765cac5345f | Running        | t3.micro      | 3/3 checks passed | us-east-1d        | ec2-3-81-107-146.com... |



Instance ID: i-07eb97765cac5345f (ubuntu\_test)

Current user data: User data currently associated with this instance.

```
#!/bin/bash

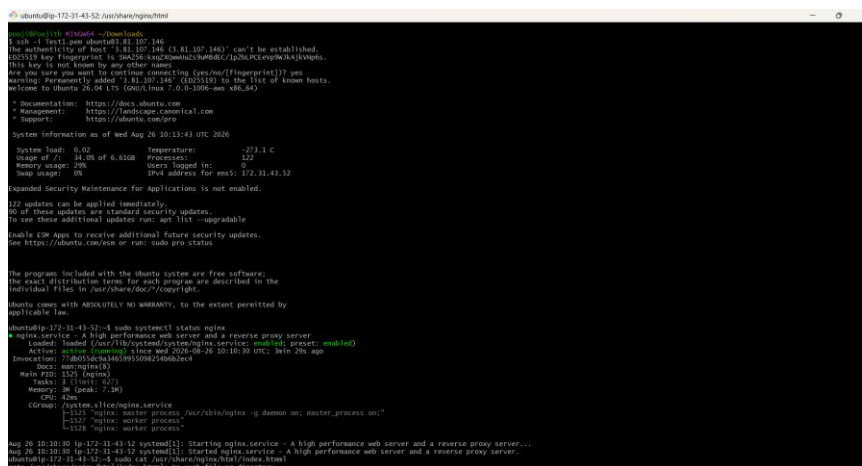
sudo apt update -y

sudo apt install nginx -y
```

Copy user data

To edit your instance's user data you first need to stop your instance.

Cancel Save



```
ubuntu@ip-172-31-43-52: ~$ cat /dev/null
cat: /dev/null: File exists

ubuntu@ip-172-31-43-52: ~$ sudo systemctl status nginx
nginx.service - A high performance web server and a reverse proxy server
Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
Active: active (running) since Wed 2020-08-26 10:10:30 UTC; 2min 29s ago
Invocation: 778d035dca3a48599550885460b2a64
Docs: man:nginx.8
Main PID: 1251 (nginx)
Tasks: 1 (limit: 22)
Memory: 36 (peak: 7.3M)
CPU: 0ms
CGroup: /system.slice/nginx.service
├─1251 nginx: master process /usr/sbin/nginx -g daemon on; master_process on;
└─1252 nginx: worker process

Aug 26 10:10:30 ip-172-31-43-52 systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server...
Aug 26 10:10:30 ip-172-31-43-52 systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server.
ubuntu@ip-172-31-43-52: ~$ sudo cat /usr/share/nginx/html/index.html
cat: /usr/share/nginx/html/index.html: No such file or directory
```

```

Aug 26 10:10:30 ip-172-31-43-52 systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server...
Aug 26 10:10:30 ip-172-31-43-52 systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server.
ubuntu@ip-172-31-43-52:~$ sudo cat /usr/share/nginx/html/index.html
cat: /usr/share/nginx/html/index.html: no such file or directory
ubuntu@ip-172-31-43-52:~$ cd /usr/share/nginx/html/
ubuntu@ip-172-31-43-52:~$ ls
index.html
ubuntu@ip-172-31-43-52:~$ cd /usr/share/nginx/html$ cat index.html
<!DOCTYPE html>
<html>
<head>
<title>welcome to nginx!</title>
<style>
html { color: #000; font-family: sans-serif; }
body { width: 35em; margin: 0 auto; }
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<div>welcome to nginx!</div>
<p>if you see this page, the nginx web server is successfully installed and
working. Further configuration is required.</p>
<p>For online documentation and support please refer to
<a href="http://nginx.org/">http://nginx.org/</a>
Commercial support is available at
<a href="http://nginx.com/">http://nginx.com/</a>
</p>
<p>Thank you for using nginx.</p>
</body>
</html>
Hello, the nginx process is installed and started using bootstrapping process
ubuntu@ip-172-31-43-52:~$ cd /usr/share/nginx/html$

```

### 3. Launch one Windows server and install Tomcat on Windows.

- ➔ Launched an EC2 instance using a Windows Server AMI.
- ➔ Configured the instance type, key pair, subnet, and security group.
- ➔ Connected to the Windows EC2 instance using Remote Desktop Protocol (RDP) from MobaXterm.
- ➔ Installed Java on the Windows server.
- ➔ Downloaded and extracted Apache Tomcat.
- ➔ Configured Tomcat and started the Tomcat server.
- ➔ Verified the Tomcat server using port 8080.

```

C:\Users\Administrator>java -version
openjdk version "21.0.12.1" 2026-08-18 LTS
OpenJDK Runtime Environment Corretto-21.0.12.9.1 (build 21.0.12.1+9-LTS)
OpenJDK 64-Bit Server VM Corretto-21.0.12.9.1 (build 21.0.12.1+9-LTS, mixed mode, sharing)

C:\Users\Administrator>javac -version
javac 21.0.12.1

C:\Users\Administrator>where java
C:\Program Files\Amazon Corretto\jdk21.0.12_9\bin\java.exe

C:\Users\Administrator>echo %JAVA_HOME%
C:\Program Files\Amazon Corretto\jdk21.0.12_9

C:\Users\Administrator>setx JAVA_HOME "C:\Program Files\Amazon Corretto\jdk21.0.12_9" /M
SUCCESS: Specified value was saved.

C:\Users\Administrator>setx PATH "%PATH%;%JAVA_HOME%\bin" /M
SUCCESS: Specified value was saved.

```

```

C:\Tomcat\apache-tomcat-9.0.121\bin>startup.bat
Using CATALINA_BASE: "C:\Tomcat\apache-tomcat-9.0.121"
Using CATALINA_HOME: "C:\Tomcat\apache-tomcat-9.0.121"
Using CATALINA_TMPDIR: "C:\Tomcat\apache-tomcat-9.0.121\temp"
Using JRE_HOME: "C:\Program Files\Amazon Corretto\jdk21.0.12_9"
Using CLASSPATH: "C:\Tomcat\apache-tomcat-9.0.121\bin\bootstrap.jar;C:\Tomcat\apache-tomcat-9.0.121\bin\tomcat-juli.jar"
Using CATALINA_OPTS: ""

C:\Tomcat\apache-tomcat-9.0.121\bin>tasklist | findstr /I java
java.exe 7948 RDP-Tcp#0 2 80,796 K

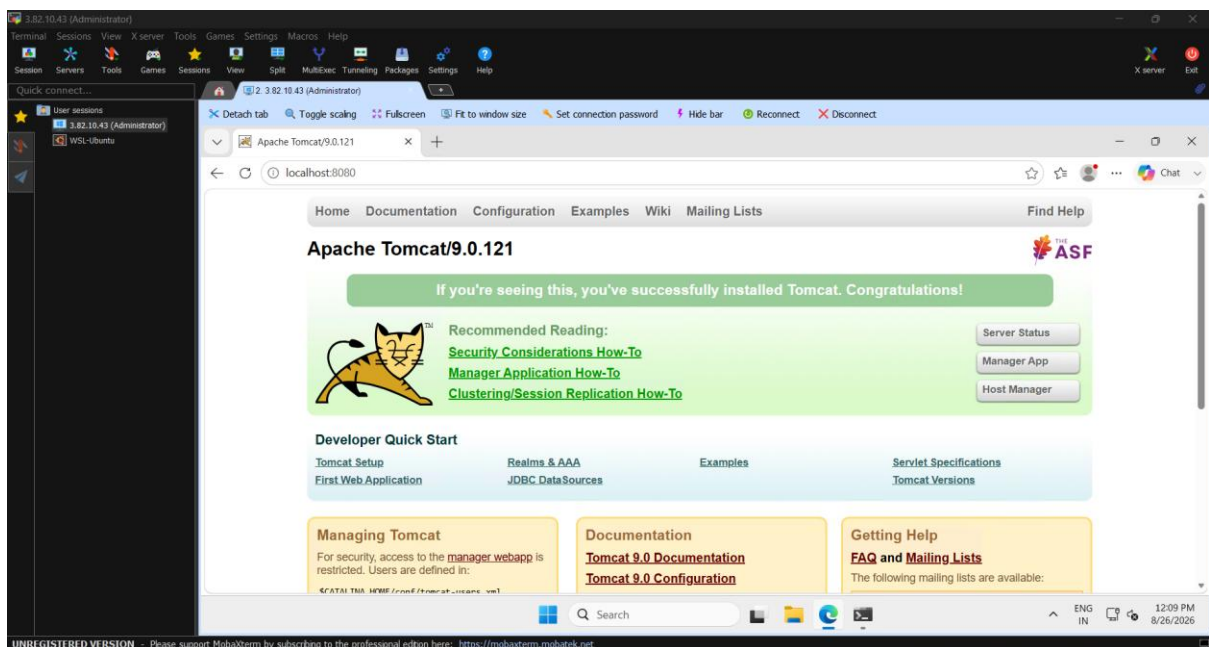
C:\Tomcat\apache-tomcat-9.0.121\bin>netstat -ano | findstr :8080
TCP 0.0.0.0:8080 0.0.0.0:0 LISTENING 7948
TCP [::]:8080 [::]:0 LISTENING 7948

```

```

Using CATALINA_OPTS: ""
C:\Tomcat\apache-tomcat-9.0.121\bin>netstat -ano | findstr :8080
TCP
TCP
TCP
TCP
26-Aug-2026 12:18:34.713 INFO [main] org.apache.coyote.AbstractProtocol.init Initializing ProtocolHandler ["http-nio-8080"]
26-Aug-2026 12:18:34.833 INFO [main] org.apache.catalina.startup.Catalina.load Server initialization in [2000] milliseconds
26-Aug-2026 12:18:35.041 INFO [main] org.apache.catalina.core.StandardService.startInternal Starting service [Catalina]
26-Aug-2026 12:18:35.041 INFO [main] org.apache.catalina.core.StandardEngine.startInternal Starting Servlet engine: [Apache Tomcat/9.0.121]
26-Aug-2026 12:18:35.067 INFO [main] org.apache.catalina.startup.HostConfig.deployDirectory Deploying web application directory [C:\Tomcat\apache-tomcat-9.0.121\webapps\docs]
26-Aug-2026 12:18:35.958 INFO [main] org.apache.catalina.startup.HostConfig.deployDirectory Deployment of web application to directory [C:\Tomcat\apache-tomcat-9.0.121\webapps\docs] has finished in [890] ms
26-Aug-2026 12:18:35.960 INFO [main] org.apache.catalina.startup.HostConfig.deployDirectory Deploying web application directory [C:\Tomcat\apache-tomcat-9.0.121\webapps\examples]
26-Aug-2026 12:18:37.131 INFO [main] org.apache.catalina.startup.HostConfig.deployDirectory Deployment of web application to directory [C:\Tomcat\apache-tomcat-9.0.121\webapps\examples] has finished in [1,171] ms
26-Aug-2026 12:18:37.133 INFO [main] org.apache.catalina.startup.HostConfig.deployDirectory Deploying web application directory [C:\Tomcat\apache-tomcat-9.0.121\webapps\host-manager]
26-Aug-2026 12:18:37.289 INFO [main] org.apache.catalina.startup.HostConfig.deployDirectory Deployment of web application to directory [C:\Tomcat\apache-tomcat-9.0.121\webapps\host-manager] has finished in [155] ms
26-Aug-2026 12:18:37.290 INFO [main] org.apache.catalina.startup.HostConfig.deployDirectory Deploying web application directory [C:\Tomcat\apache-tomcat-9.0.121\webapps\manager]
26-Aug-2026 12:18:37.446 INFO [main] org.apache.catalina.startup.HostConfig.deployDirectory Deployment of web application to directory [C:\Tomcat\apache-tomcat-9.0.121\webapps\manager] has finished in [156] ms
26-Aug-2026 12:18:37.448 INFO [main] org.apache.catalina.startup.HostConfig.deployDirectory Deploying web application directory [C:\Tomcat\apache-tomcat-9.0.121\webapps\ROOT]

```



#### 4. Take a snapshot of the instance created in Task 1.

- ➔ Identified the EC2 instance created in Task 1.
- ➔ Identified the EBS volume attached to the instance.
- ➔ Created an EBS snapshot of the attached volume.
- ➔ Added a suitable snapshot description and tags.
- ➔ Verified the snapshot in the EBS Snapshots section.

| Name | Snapshot ID            | Full snapshot size | Volume size | Description              | Storage tier | Snapshot status |
|------|------------------------|--------------------|-------------|--------------------------|--------------|-----------------|
|      | snap-047beaf3239e0af66 | -                  | 8 GiB       | Snapshot_of_amazon_image | Standard     | Pending         |

**EC2**

- Dashboard
- AWS Global View
- Events
- Instances**
  - Instances
  - Instance Types
  - Launch Templates
  - Spot Requests
  - Savings Plans
  - Reserved Instances
  - Dedicated Hosts
  - Capacity Reservations
  - Capacity Manager
- Images**
  - AMIs
  - AMI Catalog
- Elastic Block Store**
  - Volumes

**snap-047beaf3239e0af66**

Last updated less than a minute ago [Refresh] [Delete] [Actions]

| Details                                                        |                                                                                           |                                                     |                                                |
|----------------------------------------------------------------|-------------------------------------------------------------------------------------------|-----------------------------------------------------|------------------------------------------------|
| <b>Snapshot ID</b><br><a href="#">snap-047beaf3239e0af66</a>   | <b>Full snapshot size</b><br>-                                                            | <b>Progress</b><br><div style="width: 0%;">0%</div> | <b>Snapshot status</b><br><span>Pending</span> |
| <b>Owner</b><br><a href="#">956035085469</a>                   | <b>Started</b><br><a href="#">Wed Aug 26 2026 17:57:54 GMT+0530 (India Standard Time)</a> | <b>Product codes</b><br>-                           | <b>Fast snapshot restore</b><br>-              |
| <b>Description</b><br><a href="#">Snapshot_of_amazon_image</a> |                                                                                           |                                                     |                                                |
| ▼ Source volume                                                |                                                                                           |                                                     |                                                |
| <b>Volume ID</b><br><a href="#">vol-03c19ea6b8b44a493</a>      | <b>Volume size</b><br><a href="#">8 GiB</a>                                               |                                                     |                                                |
| ▼ Encryption                                                   |                                                                                           |                                                     |                                                |
| <b>Encryption</b><br>Not encrypted                             | <b>KMS key ID</b><br>-                                                                    | <b>KMS key alias</b><br>-                           | <b>KMS key ARN</b><br>-                        |
| Snapshot settings                                              |                                                                                           |                                                     |                                                |
| <b>Storage tier</b>                                            | <b>Tags</b>                                                                               |                                                     |                                                |
| ▼ Snapshot Lock                                                |                                                                                           |                                                     |                                                |

```

root@ip-172-31-43-52:~# find / -name authorized_keys
/home/ubuntu/.ssh/authorized_keys
/root/.ssh/authorized_keys
root@ip-172-31-43-52:~# vi /home/ubuntu/.ssh/authorized_keys
root@ip-172-31-43-52:~# nano /home/ubuntu/.ssh/authorized_keys
root@ip-172-31-43-52:~# exit
logout
ubuntu@ip-172-31-43-52:~$ exit
logout
Connection to 54.80.156.156 closed.

poojith@Poojith MINGW64 ~/Downloads
$ ssh ubuntu@54.80.156
welcome to ubuntu 26.04 LTS (GNU/Linux 7.0.0-1006-aws x86_64)

+ Documentation: https://docs.ubuntu.com
+ Management: https://landscape.canonical.com
+ Support: https://ubuntu.com/pro

System information as of Wed Aug 26 12:43:57 UTC 2026

System load: 0.0 Temperature: -273.1 C
Usage of /: 34.8% of 6.61GB Processes: 122
Memory usage: 34% Users logged in: 0
Swap usage: 0% IPv4 address for ens5: 172.31.43.52

Expanded Security Maintenance for Applications is not enabled.

122 updates can be applied immediately.
90 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Wed Aug 26 12:38:56 2026 from 103.203.172.230
ubuntu@ip-172-31-43-52:~$

```

## 6. Launch any EC2 using the spot purchasing option.

- ➔ Selected launch instance option
- ➔ Selected all the required elements
- ➔ Under additional settings → purchasing option → selected spot purchasing
- ➔ Launched the instance.

The screenshot shows the AWS Management Console for the 'Instances' page. The left sidebar contains navigation options like Dashboard, AWS Global View, Events, Instances, Instance Types, Launch Templates, Spot Requests, Savings Plans, Reserved Instances, Dedicated Hosts, Capacity Reservations, Capacity Manager, Images, and Elastic Block Store. The main content area displays a table of instances. One instance, 'Spot\_Instance' (ID: i-0f9394ad78b4c4946), is selected and highlighted. Below the table, the details for this instance are shown, including its state (Running), type (t3.micro), and various configuration settings like launch time, instance auto-recovery, and AMI location.

| Name                 | Instance ID                | Instance state | Instance type   | Status check        | Availability Zone | Public IPv4 DNS                 |
|----------------------|----------------------------|----------------|-----------------|---------------------|-------------------|---------------------------------|
| Amazon_Linux...      | i-0f2e33850e570f722        | Stopped        | t3.micro        | -                   | us-east-1d        | -                               |
| ubuntu_test          | i-07eb97765cac5345f        | Stopped        | t3.micro        | -                   | us-east-1d        | -                               |
| windows_tomcat       | i-074d07f55dec9eb0c        | Stopped        | t3.micro        | -                   | us-east-1d        | -                               |
| <b>Spot_Instance</b> | <b>i-0f9394ad78b4c4946</b> | <b>Running</b> | <b>t3.micro</b> | <b>Initializing</b> | <b>us-east-1d</b> | <b>ec2-100-54-246-168.co...</b> |

**i-0f9394ad78b4c4946 (Spot\_Instance)**

- Stop protection:** Disabled
- Instance reboot migration:** Default (On)
- Stop-hibernate behavior:** Disabled
- State transition reason:** -
- Launch time:** Wed Aug 26 2026 22:33:19 GMT+0530 (India Standard Time) (1 minute)
- Instance auto-recovery:** Default
- AMI Launch index:** 0
- Credit specification:** unlimited
- AMI location:** amazon/al2023-ami-2023.12.20260817.0-kernel-6.18-x86\_64
- Lifecycle:** spot
- Key pair assigned at launch:** Test1
- Kernel ID:** -

## 7. Enable termination policy on the EC2 created in Task 2.

- ➔ Selected the instance which I want to enable the termination policy
- ➔ Selected actions → instance settings → Change termination Protection
- ➔ Selected enable option in termination protection.

